Numerical Simulation of Planetary Orbits and Alignments

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1 Introduction

Planetary motion is a standard example of why numerical simulation is needed in physics. A single ideal two-body orbit can be treated analytically, but a realistic Solar System is a many-body problem in which every body perturbs every other body. Long-term orbital prediction therefore depends on numerical integration rather than closed-form solutions [2].

In this project I develop a two-dimensional object-oriented Python code to simulate the Sun together with Mercury, Venus, Earth, Mars, and Jupiter. I use the code to complete Experiment 1 on orbital periods, Experiment 2 on energy conservation and alternative integrators, and Experiment 4 on planetary alignment. My aim is to assess both the physical behaviour of the model and the numerical behaviour of the algorithms. I therefore test how accurately the simulated periods reproduce accepted sidereal values, how strongly the timestep affects the results, which integration method is most suitable for long-term orbital motion, and how sensitive "planetary alignment" is to the criterion used.

2 Methods

I organise the code into separate modules for bodies, constants, integration methods, simulation control, analysis, plotting, and file I/O. Each gravitating object is represented by a `Body` class storing its mass, colour, position, velocity, and accelerations. A central `Simulation` class stores the full system state, computes pairwise gravitational accelerations, advances time, and records trajectories and energies. The numerical update rules are isolated in three integrator classes: Beeman, Euler–Cromer, and Direct Euler. This structure lets all experiments run on one shared simulation engine rather than on separate copies of the physics code.

The simulation uses astronomical units for distance, Earth masses for mass, and years for time. With this choice of units the gravitational constant is

$$G = \frac{4\pi^2}{332\,946},$$

so that a circular orbit of radius 1 AU around the Sun has period 1 year. Planet parameters are read from a JSON input file. I place every planet initially on the positive x-axis at its orbital radius and assign it a circular-orbit speed

$$v = \sqrt{\frac{GM}{r}}$$

in the positive y-direction. I then set the Sun's initial velocity so that the total momentum is zero and shift all positions so that the initial centre of mass is at the origin. These choices make the initial state dynamically consistent, but they also simplify the real Solar System by assuming circular, coplanar, same-phase starting orbits.

For the force calculation I use symmetric pairwise updates. For each unordered pair of bodies, the displacement vector gives equal and opposite acceleration contributions proportional to Gm/r^3. This

avoids double counting, preserves Newton's third law at the discrete level, and makes the code extensible to additional bodies.

I implement the three timestepping rules given in the project brief. Beeman predicts position using the current and previous accelerations, recomputes acceleration at the new position, and then corrects the velocity [5]. Euler–Cromer updates velocity before position, which is known to improve stability compared with the basic Euler ordering for mechanical systems [6]. Direct Euler updates position before velocity. Since Beeman requires $a(t - \Delta t)$, I initialise the previous acceleration with a backward Taylor estimate. Total energy is computed as the sum of all kinetic energies and all pairwise gravitational potential energies.

I detect orbital periods automatically in heliocentric coordinates. Because every planet starts on the positive x-axis, one full orbit corresponds to the first upward crossing of that axis after $t = 0$. I detect a crossing when the heliocentric y-coordinate changes from negative to non-negative while $x > 0$, and I linearly interpolate between consecutive steps to estimate the crossing time more accurately. I compare the simulated periods with NASA sidereal values [1].

For Experiment 1 I run the Beeman simulation for 13 years, long enough to capture Jupiter's first orbit, and I then perform a timestep study from 50 to 8000 steps per Earth year. For Experiment 2 I compare Beeman, Euler–Cromer, and Direct Euler over 200 years at $\Delta t = 1/2000$ years, and I also produce a dedicated 20-year Beeman zoom to resolve the fine structure of the energy error. For Experiment 4 I run a 2000-year Beeman simulation at $\Delta t = 1/1000$ years. I define an alignment by requiring all five planets to lie within a threshold angle of the arithmetic mean heliocentric angle, rather than using a mean normalised position vector, because the project brief explicitly requires a mean-angle criterion. I use circular-data wrapping to handle the $0°/360°$ boundary correctly [4], test thresholds of $5°$, $10°$, $15°$, $20°$, and $30°$, and exclude the artificial $t = 0$ starting alignment from the reported statistics.

3 Results

Figure 1 compares the simulated and reference orbital periods. The agreement is close for all five planets, and the ordering of orbital times is reproduced correctly.

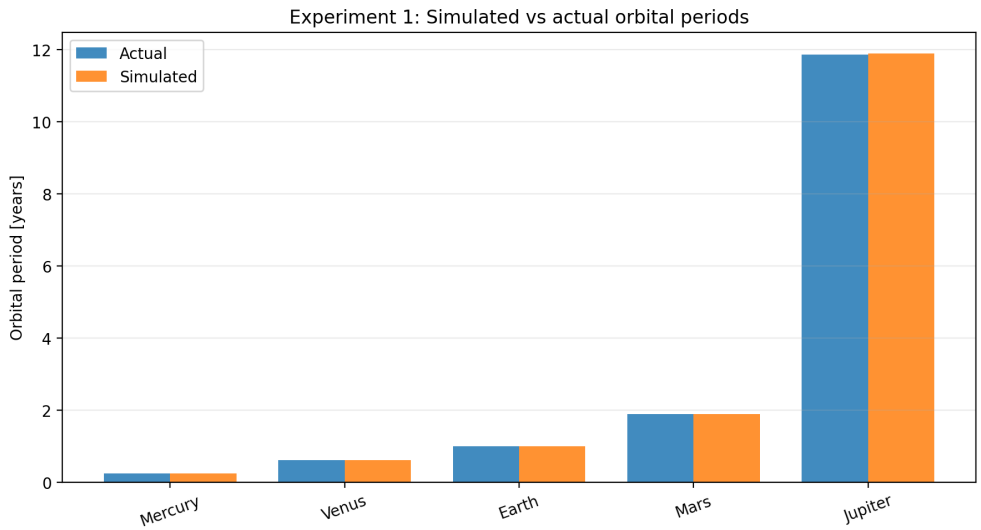

Figure 1: Experiment 1 simulated and reference sidereal orbital periods.

The exact values are listed in Table 1. All five planets are reproduced with sub-percent errors. Mercury has the smallest absolute error at 0.0738%, while Jupiter has the largest at 0.1807%. The ranking and

scale of the orbital periods are therefore reproduced correctly.

Table 1: Experiment 1 simulated orbital periods compared with NASA sidereal values [1].

Planet	Actual period [yr]	Simulated period [yr]	Error [%]
Mercury	0.2408467	0.2410243	0.0738
Venus	0.6151973	0.6158213	0.1014
Earth	1.0000000	1.0012536	0.1254
Mars	1.8808158	1.8835428	0.1450
Jupiter	11.8626150	11.8840460	0.1807

The timestep study is summarised in Figure 2. Two trends appear. First, the difference relative to the finest tested timestep decreases steadily as Δt is reduced, showing numerical convergence. Second, the error relative to the actual Solar System drops quickly at coarse timesteps but then plateaus near 0.1%. The best direct match to the NASA reference values occurs at $\Delta t = 0.003333$ years (300 steps per Earth year), where the mean absolute period error is 0.1147% and the maximum error is 0.1806%. However, this reflects partial cancellation between numerical and modelling errors rather than a fundamentally better timestep. A more reliable convergence criterion is comparison with the finest tested run. Under that measure, $\Delta t = 0.002$ years (500 steps per year) is already within 0.05% of the 8000-step reference, and the default choice $\Delta t = 0.0005$ years lies safely inside the converged regime.

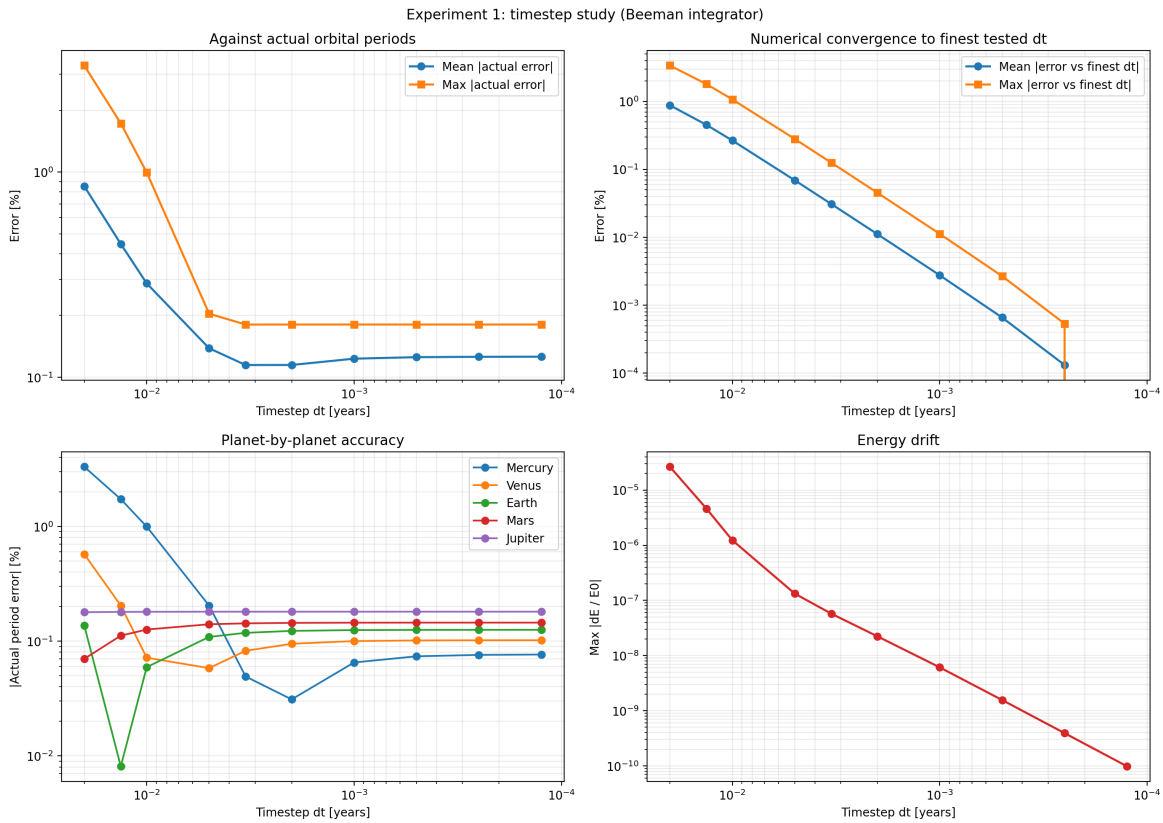

Figure 2: Combined Experiment 1 timestep study. The panels show error against the real Solar System, error against the finest tested timestep, planet-by-planet accuracy, and maximum relative energy drift.

Experiment 2 shows a much clearer separation between algorithms. Table 2 and Figure 3 show that Beeman has the smallest energy drift, Euler–Cromer remains bounded but less accurate, and Direct

Euler performs very poorly. Over 200 years, Beeman has a maximum absolute relative energy drift of 1.67×10^{-9}, Euler–Cromer reaches 1.29×10^{-6}, and Direct Euler reaches 7.47×10^{-2}.

Table 2: Experiment 2 energy-drift summary over 200 years at $\Delta t = 1/2000$ years.

Method	Final $(E - E_0)/\lvert E_0\rvert$	Max $\lvert E - E_0\rvert/\lvert E_0\rvert$
Beeman	$-6.700\,000 \times 10^{-10}$	$1.670\,000 \times 10^{-9}$
Euler–Cromer	$2.240\,000 \times 10^{-7}$	$1.290\,000 \times 10^{-6}$
Direct Euler	$7.470\,000 \times 10^{-2}$	$7.470\,000 \times 10^{-2}$

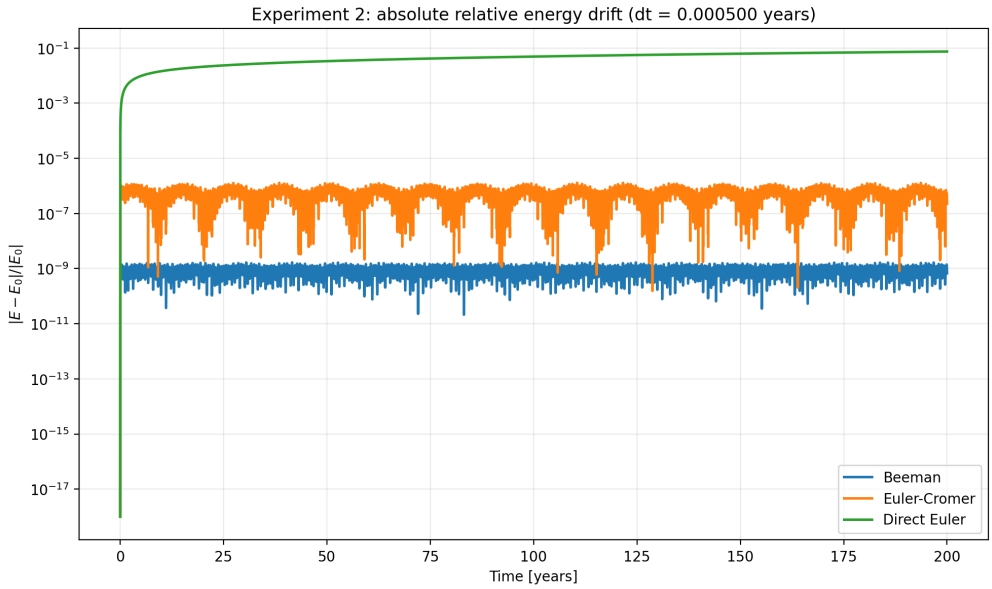

Figure 3: Experiment 2 relative total-energy drift over 200 years. Beeman performs best, Euler–Cromer is worse but bounded, and Direct Euler shows strong secular growth.

The Beeman zoom in Figure 4 shows small oscillatory deviations of order 10^{-9} rather than monotonic growth.

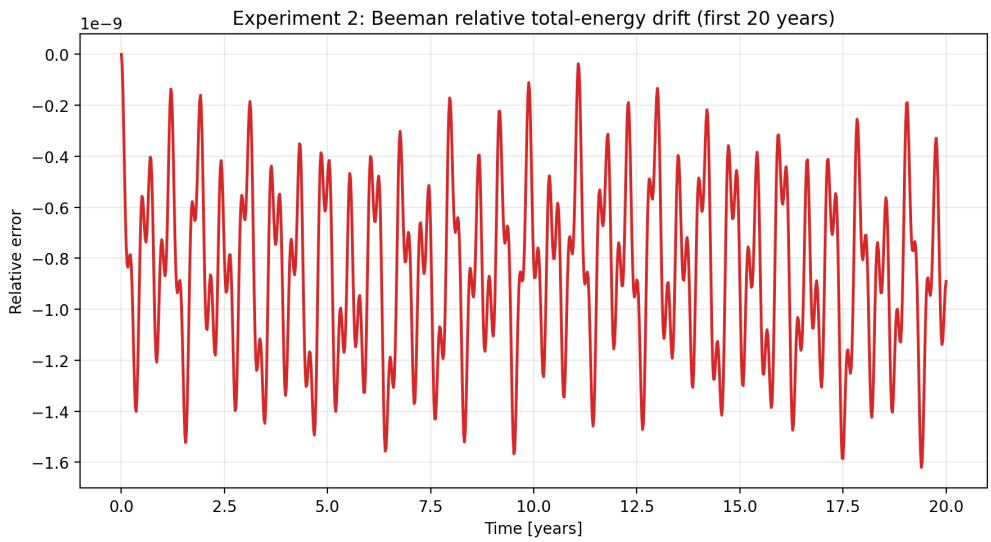

Figure 4: Beeman relative total-energy drift over the first 20 years.

Experiment 4 shows that strict five-planet alignments are extremely rare in this simplified model. At the 5° threshold, only one non-trivial event is detected in 2000 years. Its peak occurs at $t = 998.235$ years, with a maximum deviation of 4.10°, and the total event duration is 0.0040 years, about 1.46 days. This explains why the 5-degree case has no interval distribution. Table 3 shows that the event count rises rapidly as the criterion is relaxed. The mean interval between events is 217.1 years at 10°, 135.0 years at 15°, 63.0 years at 20°, and 15.8 years at 30°.

Table 3: Experiment 4 threshold summary. One non-trivial 5-degree alignment is found; looser criteria produce many more events with much shorter mean intervals.

Threshold [deg]	Events	Mean interval [yr]	Min interval [yr]	Max interval [yr]
5	1	—	—	—
10	8	217.106	44.777	476.728
15	14	134.993	38.382	279.946
20	32	62.958	6.393	196.781
30	125	15.791	1.901	44.775

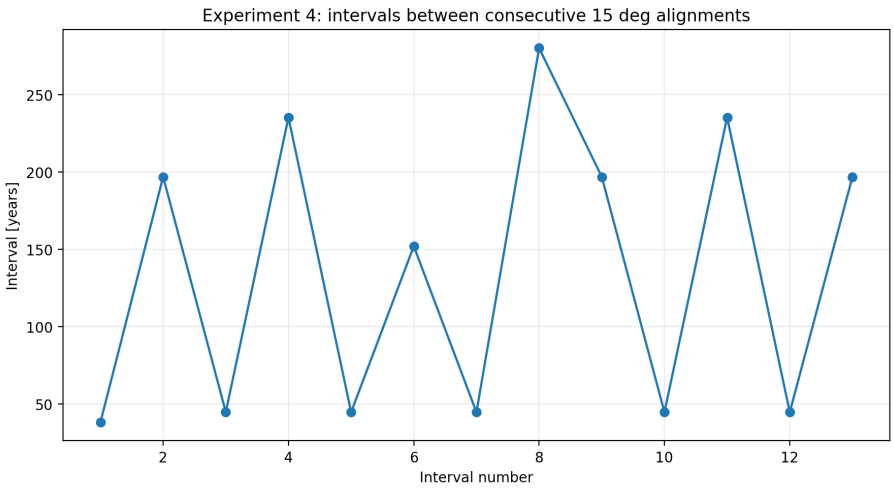

Figure 5: Intervals between consecutive 15-degree alignment events. The spacing is clearly irregular rather than periodic with a single fixed timescale.

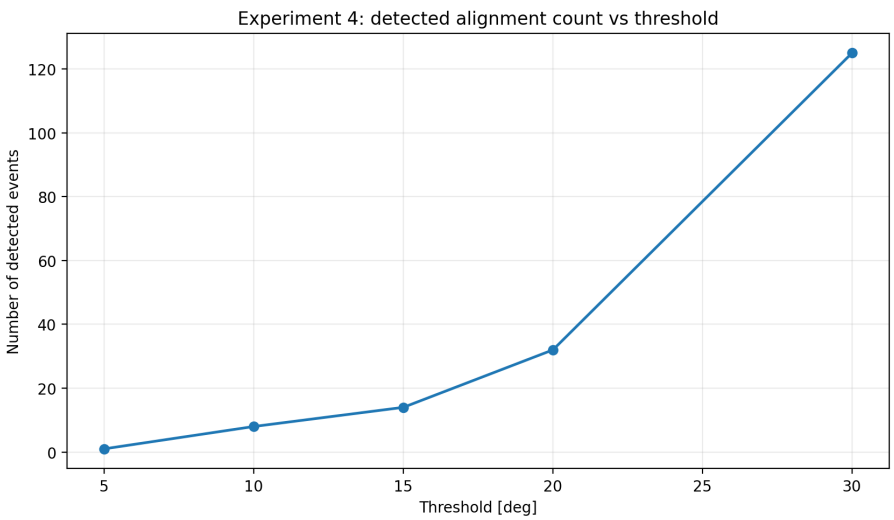

Figure 6: Detected alignment count as a function of threshold angle for Experiment 4.

4 Discussion

The experiments show that the code is physically reasonable but also clearly idealised. The period results demonstrate that the many-body implementation and Beeman integrator are working properly. At the same time, the timestep study shows that the remaining mismatch with NASA values is not purely a timestep problem. Once Δt is small enough, further reduction gives very little improvement against the real Solar System because the dominant limitations are the modelling assumptions: circular starting velocities, shared initial phase, a two-dimensional geometry, and omission of the outer planets beyond Jupiter. This is why the best match to the reference data occurs at a coarser timestep than the one ultimately adopted.

The timestep study also justifies retaining $\Delta t = 0.0005$ years for the main runs. It is not the timestep that happens to give the smallest direct error against the real Solar System, but it is comfortably inside the converged regime and gives excellent long-term energy behaviour. That makes it a stronger scientific choice than selecting a timestep purely because it benefits from accidental error cancellation.

Experiment 2 provides the clearest lesson. Direct Euler fails because it does not preserve the structure of orbital dynamics. Its energy drift is not a small calibration issue; it changes the simulated system by injecting spurious energy. Euler–Cromer is much better because its update order improves stability, but Beeman remains the best choice. This ranking is consistent with the wider literature on long-time integration of Hamiltonian systems [3]. The important point is that numerical "energy conservation" means bounded, very small variation, not a mathematically exact constant value.

The alignment experiment illustrates a different issue: the definition of the measurement is part of the science. In ordinary language, "planetary alignment" sounds like a single obvious event, but the results here depend strongly on the chosen threshold. Under a strict 5° mean-angle criterion, the event is rare enough to appear only once in 2000 years. Under looser thresholds, alignments become much more common, but their intervals are irregular, as shown directly by Figure 5. This means any claim about how often alignments occur is incomplete unless the criterion is stated explicitly. It also justifies the circular-angle treatment in the code, because naive linear averages would be misleading [4].

The most important improvement for future work is to replace the circular approximate initial conditions with ephemeris-based positions and velocities. That would reduce the residual period errors and remove the artificial shared starting geometry. A second improvement would be to extend the code to three dimensions so that orbital inclination is included. A third would be to add the remaining outer planets and test how much the alignment statistics depend on the chosen planetary subset. The modular code structure would also support a future extension to Experiment 3, in which a spacecraft body is launched from Earth towards Mars.

5 Conclusion

I build a reusable many-body Solar-System simulation and use it to complete Experiments 1, 2, and 4. The Beeman integrator reproduces the orbital periods of Mercury, Venus, Earth, Mars, and Jupiter with sub-percent errors. The timestep study shows clear numerical convergence, but it also shows that the remaining disagreement with the real Solar System is mainly caused by modelling assumptions once the timestep is sufficiently small. The energy comparison demonstrates that Beeman is the most suitable of the three tested methods, Euler–Cromer is acceptable but less accurate, and Direct Euler is unsuitable for long-term orbital simulation because of its strong secular drift. Finally, five-planet alignments are extremely sensitive to the alignment threshold: at 5° only one non-trivial event appears in 2000 years, whereas looser thresholds produce many more events with irregular spacing.

Overall, the project succeeds both physically and numerically. The core gravitational model is simple, but the experiments show that algorithm choice, timestep control, and careful analysis criteria are all crucial to obtaining meaningful results.

Word count (excluding title, equations, figures, tables, references, and appendices): 1933

References

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